MILITARY SPECIFICATION

HEATERS, SPACE: MULTIFUEL, WITH BLOWER, 15,000 BTU/HR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers liquid-multifuel-burning space heaters having 15,000 Btu/hr capacity.

1.2 Classification. The heaters shall be of the following types, as specified (see 6.2):

Type I - 120-volt ac, 50/60-Hertz, single phase.
Type II - 120-volt ac, 400-Hertz, single phase.
Type IV - 28-volt dc.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 4520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

CC-M-1807	- Motor, Alternating-Current Fractional and Integral Horsepower (500 HP and Smaller).
QQ-S-781	- Strapping, Steel, and Seals.
VV-F-800	- Fuel, Oil, Diesel.
VV-G-76	- Gasoline, Automotive.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-W-5086	- Wire, Electric, Hooking and Interconnecting Polyvinyl Chloride-Insulated, Copper or Copper Alloy Conductor.
MIL-F-5591	- Fasteners; Panel.
MIL-R-6106	- Relays, Electric, General Specification for.
MIL-B-7883	- Brazing for Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-H-10868	- Hose Assembly, Rubber, Utility, Gasoline (1/4-Inch Inside Diameter).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-R-11050	- Rectifiers, Metallic, Selenium.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-W-16878	- Wire, Electrical, Insulated, High Temperature.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

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STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-882	- System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS3057	- Clamps, Cable, Electric Conductor.
MS3100	- Connector, Receptacle, Electric, Wall Mounting, Solder Contacts, AN Type.
MS3108	- Connectors, Plug, Electric, Solder Contact, 90 Degree, AN Type.
MS51085	- Filter, Fluid, Pressure; Automotive Fuel (10 GPH Filtration).
MS51086	- Strainer, Sediment, Automotive Fuel (10 GPH).
MS51321	- Pump, Fuel, Electrical - 24 Volts D. C., 25 G.P.H. Capacity.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualification.
Power Test Code.

Supplement on Instrument and Apparatus, Part 10 - Flue Exhaust Gas Analysis.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AIR MOVING AND CONDITIONING ASSOCIATION

Bulletin No. 210 - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 West Touhy Avenue, Park Ridge, IL 60068.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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AMERICAN WELDING SOCIETY, INC.

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS

ASHRAE Handbook and Product Directory - 1977 Fundamentals.

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D2156 - Smoke Density in the Flue Gases from Distillate Fuels, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The heater shall be as shown in Figures 1 through 4 and as specified herein. The heater shall be liquid-multifuel burning and shall deliver uncontaminated, forced air. The heater shall have a measured output heating capacity of not less than 15,000 British thermal units per hour (Btu/hr) at sea level and an indicated output heating capacity of not less than 13,000 Btu/hr at 10,000-foot altitude at an ambient temperature of 70 degrees Fahrenheit (°F) plus or minus 10° F, and shall deliver airflow as specified herein. The heater shall consist of a fuel system, combustion-air blower, burner head, combination combustion chamber and heat exchanger, recirculating-air blower, all necessary controls and wiring, sheet-metal enclosure, and accessories. The weight of the heater with all its components installed shall be not more than 40 pounds. The Types I, II, and IV heaters shall operate from the following power sources, respectively: 120-volt, 50/60-Hertz, single-phase ac; 120-volt, 400-Hertz, single-phase ac; and 28-volt dc. On Type I heaters, the specified capacities shall be when operating on 60-Hertz power; a 15 percent reduction in capacity shall be allowed when operating on 50-Hertz power. Alternating current components shall be utilized for Types I and II heaters, except that the fuel pump and control system may be direct current.

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3.2 First article (preproduction model). The contractor shall furnish four heaters for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce heaters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more heaters (see 6.2), for inspection as specified in 4.4.

3.4 Electrical characteristics. Each type heater shall be capable of starting and operating on the electrical power input characteristics as follows:

	Voltage	Frequency	Voltage	Frequency
Type I				
	Rated	Rated	120	60
	High	Rated	132	60
	Low	Rated	108	60
	High	High	126	63
	Low	Low	114	48
	Low	High	114	63
	High	Low	126	48
Type II				
	Rated	Rated	120	400
	High	Rated	132	400
	Low	Rated	108	400
	High	High	126	420
	Low	Low	114	380
	Low	High	114	420
	High	Low	126	380
Type IV				
	Rated	DC	28	DC
	Low	DC	22	DC
	High	DC	30	DC

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3.5 Maintainability. The heater shall be designed to permit accessibility to all parts for servicing, maintenance, and repair utilizing only tools associated with items of this nature. Replacement and adjustment of parts shall be accomplished with minimum drainage of fuel and with minimum interference from, injury to, or removal of other parts or wiring. Ease-of-maintenance provisions shall incorporate, to the maximum extent practicable, features insuring operating clearances and facilitating applicable maintenance and servicing operations at low temperatures. Covers or plates that must be removed for adjustment or repair shall be equipped with captive, quick-disconnect fasteners. Service adjustments required shall be the minimum consistent with required performance. Accessible adjustments shall be quickly returnable to their original settings. Design of the heater shall eliminate where possible the assembly of wrong combinations of parts and shall eliminate or minimize the holding of heavy or awkward parts while being brought into final position and while being fastened. Minimum allowable time between scheduled organizational maintenance services shall be 125 hours. The mean-time-to-repair shall be 2 hours at the organizational level, 8 hours direct support and 24 hours at the general support level.

3.6 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials subject to atmospheric corrosion or rust shall not be treated with a temporary treatment such as grease or oil in order to meet environmental condition requirements. Materials shall be free from defects or imperfections which might affect the serviceability or appearance of the finished heater (see 6.10).

3.6.1 Dissimilar metals. Joints (including fasteners) between dissimilar metals in direct contact shall be avoided unless protected against galvanic corrosion. Dissimilar metals are defined in MIL-STD-889, Type II. Protection against galvanic corrosion may be accomplished by separation of dissimilar metals by means of zinc chromate tape or plating.

3.7 Fasteners. All steel fasteners shall be plated with cadmium, zinc, or nickel to resist atmospheric corrosion under the environmental conditions specified herein. Sheet metal or self-tapping screws shall not be used. Threads shall conform to Handbook H28.

3.8 Environmental conditions. The heater shall exhibit no malfunctioning, leaks, or flooding, corrosion, rough operation, ignition combustion difficulties such as sputtering, overheating, or other irregular operation when subjected to the following environmental conditions:

(a) Operation in any temperature from plus 80° F to minus 50° F and at any elevation from sea level to 10,000 feet.

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(b) Storage at an ambient temperature of plus 155° F for periods up to 4 hours daily, no solar radiation; and at minus 65° F for periods up to 72 hours, no solar radiation.

3.8.1 Shock and vibration. The heater shall withstand shock and vibration as specified herein without structural damage, visual misalinement of components, malfunctioning of components or controls, leaks, abnormal vibration, or any other irregular operation.

3.8.1.1 Shock. The heater shall withstand three shocks applied along each of the three mutually perpendicular directions that are respectively parallel to the edge of the heater (a total of 15 shocks). Each shock shall be not less than 15 g's (g = acceleration caused by gravity) and shall have a duration of not less than 11 milliseconds. The shock wave form shall be as required by Figure 516.2-2 of MIL-STD-810.

3.8.1.2 Vibration. The heater shall withstand vibration as specified in MIL-STD-810, Method 514.2, Category F, Procedure VIII, Table 514.2-VI, Curve V. The test time shall be 60 minutes total per plane.

3.8.2 Salt fog and humidity. When subjected to the tests specified herein, the heater shall show no evidence of peeling of paint, malfunctioning, rough operation, or electrical shorts and shall show no evidence of deterioration, corrosion, or change in tolerance limits of any internal or external parts which could prevent the unit from meeting operational and maintainability requirements.

3.8.2.1 Salt fog. The heater shall withstand salt-fog exposure as specified in MIL-STD-810, Method 509.1, Procedure I.

3.8.2.2 Humidity. The heater shall withstand humidity exposure as specified in MIL-STD-810, Method 507.1, Procedure I, except that the maximum temperature shall be 155° F.

3.9 Durability. The heater shall operate continually for not less than 2000 hours with the heat output varying not more than 10 percent of that obtained in the capacity test (see 4.6.2.1), run prior to the durability test (see 4.6.2.4). Except for refueling, the heater shall operate continuously for not less than 250 hours. At the end of 250 hours, cleaning of the burner head, cleaning of the fuel filter(s) and cleaning of the ignition plug(s) shall be permitted. The ignition plug(s) and fuel filter(s) may be replaced every 1000 hours. Motor brushes shall be capable of operating for not less than 800 hours on Type IV heaters, and not less than 1750 hours on Type I and Type II heaters, if applicable, without replacement.

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3.10 Tilted operation. The heater shall start and operate without failure or malfunction when positioned at any angle from horizontal to 10 degrees from horizontal in any plane.

3.11 Efficiency. The heater shall have an indicated efficiency of not less than 70 percent and shall have a measured efficiency of not less than 65 percent. The indicated efficiency shall be based on the flue gas analysis, and the measured efficiency shall be based on the measured output.

3.12 Safety. All exposed parts that are subject to high operating temperatures or that are energized electrically shall be insulated, fully enclosed, guarded, or grounded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions. Safety standards shall conform to MIL-STD-882.

3.13 Sound level. The sound level of the heaters shall conform to MIL-STD-1474, Category E for steady-state noise.

3.14 Fuel system. The heater shall operate with gasoline conforming to MIL-G-3056 or VV-G-76 or with any other gasoline (leaded or aromatic) up to 100 octane grade, and with diesel fuel conforming to VV-F-800, Class DF-1, DF-2, or DF-A, having a cloud point not lower than minus 55° F. The fuel system shall include a fuel pump conforming to MS51321. A fuel-line connection shall be mounted through the heater case and shall terminate with a brass fitting sized and externally threaded to receive a fuel hose conforming to MIL-H-10868. A filter conforming to MS51085 or MS51086 shall be provided within the heater case to filter all fuel. High-pressure atomizing nozzles shall be equipped with a fuel filter or strainer in addition to the above filter. The fuel system shall not leak.

3.14.1 Fuel-pump assembly. The pump assembly shall be located inside the heater cabinet and shall be constructed so that it may be removed and replaced as an assembly. The fuel pump shall maintain a constant fuel flow to the burner for the heater to operate at maximum output when the fuel supply is 7 feet below the fuel pump. Transformers and rectifiers may be used for supplying power to the pump. Pump operation shall be controlled by the thermostat.

3.15 Ignition system. The ignition system shall ignite the fuel-air mixture and maintain combustion at the conditions specified in 3.8. The ignition system shall be of an electrical arc type with ignition electrodes located at the fuel nozzle. Power shall be furnished for the electrodes from an ignition transformer or a solid-state electronic circuit.

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3.16 Combustion-air supply. The combustion-air blower motor shall be dripproof and fungusproof and shall conform to CC-M-1807. The blower shall be of a size and type which will supply air to the burner at a pressure for efficient operation of the burner at specified output. The motor shall be equipped with lubricated-for-life, sealed ball bearings.

3.17 Burner. The burner shall be of a design that is not critical in regard to the fuel-air ratio required for maintaining continuous combustion for all conditions specified in 3.8 and 3.14. Operation on any required fuel shall produce smoke not in excess of Bacharach No. 4 in the flue gases and efficiency not less than specified in 3.11. The burner shall not become clogged by the products of combustion. The burner shall be located and attached to the combustion chamber so as to facilitate removal of the burner components for inspection, replacement, and adjustment. Burner operation shall not be hindered by the back pressure induced by a flue-gas exhaust pipe equivalent to the resistance of four 1-1/4-inch, 90-degree elbows and 6 feet of 1-1/4-inch diameter pipe.

3.18 Combustion chamber and heat exchanger. The combustion chamber and heat exchanger shall be constructed of steel. The primary pass of the heat exchanger shall serve both as the housing for the burner and as the combustion chamber. When used to extract heat, baffles or fins shall be continuously bonded to the heat exchanger by methods that will withstand the temperatures encountered. The heat exchanger and the combustion chamber shall be sealed to prevent the products of combustion from contaminating the circulated air. The exhaust fitting shall be of the same material as the exchanger.

3.19 Circulation-air blower. The circulation-air blower shall be motor driven and shall be located within the heater case. The blower, without discharge-air grille (see 3.21.1), shall deliver air at not less than the following rates:

(a) 115 standard cubic feet per minute (scfm) against an external back pressure of 0.20 inch water at sea level conditions (0 to 500 feet).
(b) 100 scfm against an external back pressure of 0.35 inch water at sea level conditions (0 to 500 feet).

The heater may draw return air from either the top or side of the heater. Blower operation shall be continued after each heating cycle to purge the heater. The motor shall be dripproof and fungusproof and shall conform to CC-M-1807. The same motor may be used for driving this blower and the combustion-air blower. The motor shall be equipped with lubricated-for-life sealed ball bearings. All references to airflow specified herein are based on standard air (air at 70° F and 0.075 pound per cubic foot density,

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and 29.92 inches of mercury). The motor shall be equipped with a thermal-overload protector to prevent motor operation above the safe operating temperature for which the motor is rated. Brush-type motors shall be provided with accessible access opening for the inspection or renewal of brushes and springs, or shall be capable of operating for the entire 2000 hour durability requirement (see 3.9), without brush maintenance or replacement.

3.20 Electrical system and controls. All internal wires shall conform to MIL-W-5086 or MIL-W-16878, Type B, except where required to be otherwise for high-voltage spark wires, transformer leads, and electromagnetic interference. Any selenium rectifiers employed shall conform to MIL-R-11050. The electrical system and heater case shall be protected by a ground circuit connected to the ground terminal of the power-supply receptacle. Electric contact points and relays shall be hermetically sealed and shall conform to MIL-R-6106.

3.20.1 Temperature control. A room thermostat shall be provided for remote mounting from the heater in the space being heated. The thermostat shall control the heater operation within plus or minus 1-1/2° F of the dial setting. The thermostat shall be of the heavy-duty type designed to withstand frequent cycling (15 to 20 cycles per hour) and shall operate in any position. The temperature range shall be from plus 55° F or less to plus 85° F or more, and the selector dial shall be so calibrated. The thermostat shall be metal encased and all nonmetal parts shall be protected against breakage.

3.20.2 On-off switch. A toggle switch shall provide on-off control of the heater. The switch shall be mounted so that the "ON" position is up.

3.20.3 Safety controls.

3.20.3.1 High-temperature safety. A nonadjustable automatic control shall be provided to shutoff the fuel supply to the combustion chamber in the event of air temperatures detrimental to components within the heater or before the air discharge temperature exceeds 275° F. The control shall be an automatic reset type.

3.20.3.2 Flame-failure safety. An automatic control shall be provided to shutoff the fuel supply to the combustion chamber and stop all heater operation in the event of either ignition failure or flame failure. The fuel shutoff shall be accomplished within 60 seconds after failure has occurred. A manual reset shall be provided.

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3.20.4 Receptacles.

3.20.4.1 Power-supply receptacle. A 3-contact, power receptacle conforming to MS3100, part number MS3100R-14S-7P, with cap and chain, shall be flush mounted in the heater case. A mating plug conforming to MS3108, part number MS3108R-14S-7S, and a cable clamp conforming to MS3057 with weatherproof sleeve shall be furnished.

3.20.4.2 Thermostat receptacle. A 3-contact, power receptacle conforming to MS3100, part number MS3100R-14S-7SX, with cap and chain, shall be flush mounted in the heater case. A mating plug conforming to MS3108, part number MS3108R-14S-7PX, with a cable clamp conforming to MS3057 and weatherproof sleeve shall be furnished.

3.20.5 Electrical protective devices. Circuit breakers shall be provided in the electrical system to protect the heaters from damage caused by maladjustment of the controls, short or open circuits, or failure of circuit parts. The circuit breakers shall be accessible from the control panel. A fuse may be used for the control circuit and may be mounted in the control panel. When a fuse is utilized, a replacement fuse shall be mounted in the heater case and shall be accessible from the control panel. All motor(s) shall be equipped with automatic-resetting, thermal-overload protectors sized to prevent motor operation above the safe operating temperature for which the motor is rated.

3.21 Heater case. The heater case shall be constructed of sheet metal having a nominal thickness of 0.0478 inch. All removable panels and access doors shall be provided with fasteners conforming to MIL-F-5591, Type II, class and style optional. The top panel and the left side panel of the heater (as designated when facing the front of the heater) shall be interchangeable. When specified (see 6.2), an additional accessory panel shall be furnished with transition and adaptation for connecting a 6 inch (inside diameter) flexible duct. When furnished, this panel shall be in addition to and interchangeable with the top and left side panels. Provisions shall be made to require the interchangeable panels to be mounted in one position only with regard to the front of the heater. Four captive nuts shall be provided on the heater base for mounting purposes. The power receptacles shall be positioned not less than 3 inches apart (center to center of the receptacles) so as to allow placement and removal of plug adapters by an operator wearing protective gloves.

3.21.1 Discharge-air grille. A grille shall be mounted on the air discharge from the heater. The grille shall be manually adjustable to direct the discharge air either up, down, left, or right. The airflow resistance of the grille shall be not more than 0.20 inch water with an airflow rate of 115 cfm.

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3.21.2 Air intake screen. A screen to prevent the entrance of foreign matter into the heater shall be mounted in the air intake opening of the heater. Screen openings shall be of a size to prevent the entrance of a 3/4-inch-diameter sphere.

3.22 Electromagnetic interference. The electromagnetic interference emission characteristics of the heaters shall conform to MIL-STD-461A, Notice 4. Specific subtests of MIL-STD-461A shall be RE02 and CE03.

3.23 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.23.1 Lubricants. The heater shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.23.2 Grease lubrication. All grease lubrication, including lubrication of sealed bearings shall be with greases as follows:

(a) MIL-G-23827 for instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 for bearings where operating temperatures do not exceed 225° F.

The heaters shall be assembled, tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.24 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.24.1, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.24.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102 - Southern California Air Pollution Control District.

3.25 Identification marking. The heater shall be identified in accordance with MIL-STD-130. The marking shall be applied to the heater on identification plates conforming to MIL-P-15024/6, Type H, Style III, Size 12.

3.25.1 Other markings. The heater shall be equipped with markings identifying items that are involved in its operation and handling such as controls, fuses, and input power connections.

3.26 Instruction plates. Each heater shall be equipped with instruction plates including warnings, diagrams, and cautions conspicuously describing any special or important procedures to be followed in operating and servicing the heater. The instruction plates shall conform to MIL-P-514, Type III, Composition C, Class 1. The words "Warning" and "Caution" shall be red, matching color No. 31136 of FED. STD. No. 595.

3.26.1 Schematic and wiring diagram. A schematic diagram and a connection (wiring) diagram shall be mounted on the inside of a removable panel of the heater. The diagram shall portray the complete operational and functional sequence of the circuit for analysis and maintenance of the heater. The connection (wiring) diagram shall portray the physical location of all wiring interconnections in the same relationship as when they are installed. Identification markings of wires, instruments, control devices, circuit-breakers, fuses, and connections shall be shown in both diagrams.

3.26.2 Selenium rectifier caution plate. When selenium rectifiers (see 3.20) are used, a caution plate shall be conspicuously located adjacent to each rectifier or group of rectifiers and shall state the following: "If malfunctioning of the rectifier occurs, do not handle the damaged rectifier while it is hot. To avoid inhalation of the poisonous fumes or absorption of poisonous selenium compound through the skin, the rectifier compartment shall be ventilated thoroughly before the rectifier chassis is removed."

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3.26.3 Power supply warning plate. A warning plate shall be placed adjacent to the power-supply receptical instructing personnel to disconnect power supply before performing maintenance.

3.27 Treatment and painting. The portions of the heater normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2).

3.28 Stenciling. The weight of each heater shall be legibly stenciled on both sides of the heater case in numerals not less than 1 inch in height. All stenciling shall conform to MIL-T-704.

3.29 Human factors. The human engineering characteristics of the heater shall provide for operation by personnel in all clothing and shall be in accordance with MIL-STD-1472.

3.30 Reliability. The specified mean-time-between-failure (MTBF) shall be 480 hours when tested as specified in 4.6.2.13. The lower test MTBF shall be 320 hours.

3.31 Workmanship.

3.31.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except where sharp edges or corners are required or where they are not detrimental to safety.

3.31.2 Brazing. Brazing of steels, copper alloys and nickel alloys shall conform to MIL-B-7883.

3.31.3 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.31.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

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3.31.5 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.31.6 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the heater to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the heater shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.1.2 Acceptability criteria. Heaters which conform to all requirements in Sections 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction heaters shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction heaters shall be subjected to the tests marked "X" in Column 2 of Table I. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production heaters will be selected at random by the Government from the heaters being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 4 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production heater shall not exclude the remaining heaters from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of any initial production heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production heaters and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on heaters previously accepted and

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produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all heaters accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each heater shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each heater shall be subjected to the test marked "X" in Column 3 of Table I. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The heater shall be examined as specified herein, for the following defects:

101. Weight not as specified.
102. Dimensions not as specified.
103. Components, other than the fuel connection and the exhaust pipe, extend beyond the dimensional limits of the heater case.
104. Ease of maintenance provisions not as specified.
105. Material not as specified.
106. Dissimilar metals not isolated or not treated as specified.
107. Fasteners not as specified.
108. Parts hazardous to operating or maintenance personnel not insulated, fully enclosed, guarded, or grounded.
109. Parts or components not as specified or missing.
110. Fuel system leaks.
111. Mating plug that fits the power receptacle not furnished.
112. Mating plug that fits the thermostat receptacle not furnished.
113. Identification or instruction markings not as specified or missing.
114. Identification plates missing or not located as specified.
115. Instruction plates missing or not as specified.
116. Treatment and painting not as specified.
117. Workmanship not as specified.
118. Electromagnetic interference suppression components and network not identical with approved model.
119. Lubrication tag(s) improperly located or missing.

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4.6.2 Tests. The test schedule is contained in Table I. Tests shall be conducted at prevailing room ambient temperatures and employing 60 Hertz power for Type I unless otherwise specified herein. All components and accessories shall be utilized in all tests unless otherwise specified herein. All operating tests shall be conducted with the heater burning the same type of gasoline and diesel fuel specified in 3.14. The heater shall be lubricated with military lubricants prior to testing. All instruments shall be calibrated prior to tests.

Table I. Test Schedule 1/ 3/

Type of inspection						
Inspection Sequence	Preproduction 4/	Quality Conformance	Initial Production	Test	Inspection paragraphs	Requirement paragraph
1	2	3	4	5	6	7
1 *	X 2/	-	-	Pressure.	4.6.2.9	3.1, 3.18, 3.30
2 *	X	-	X	Controls.	4.6.2.3	3.20
3 *	X	-	X	Capacity and airflow.	4.6.2.1	3.1, 3.11, 3.17, 3.19, 3.21.1, 3.21.2
4	X	-	X	Sound-level.	4.6.2.5	3.13
5	X	-	X	Tilted-operation.	4.6.2.6	3.10
6	X	-	-	High-temperature.	4.6.2.2.3	3.8
7	X	-	-	Altitude.	4.6.2.2.7	3.1
8	X	-	X	Low-temperature.	4.6.2.2.4	3.8
9	X	-	X	Humidity.	4.6.2.2.1	3.8
10	X	-	-	Salt-spray.	4.6.2.2.2	3.8
11	X	-	X	Vibration.	4.6.2.2.5	3.8.1
12	X	-	-	Shock.	4.6.2.2.6	3.8.1
13	X	-	-	Controls.	4.6.2.3	3.20
14 *	X	-	-	Durability.	4.6.2.4	3.9
15	X	-	X	Electromagnetic interference.	4.6.2.7	3.22
16 *	X	-	-	Pressure.	4.6.2.9	3.30
17	-	X	-	Individual performance.	4.6.2.8	3.1, 3.2
18	X	-	-	Human factors.	4.6.2.11	3.29

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Table I. Test Schedule 1/ 3/ (Cont'd)

1	2	3	4	5	6	7
19	-	-	X	Reliability.	4.6.2.13	3.30
20	-	-	X	Maintenance evaluation.	4.6.2.12	3.5
21	-	-	X	Safety.	4.6.2.10	3.12

1/ Perform inspection marked "X".
2/ May be performed prior to assembly.
3/ The tests shall be performed in the sequence indicated, or as approved by the contracting officer.
4/ Three preproduction heaters shall be subjected to tests indicated by *. All other tests shall be performed on the fourth preproduction heater.

4.6.2.1 Capacity and airflow. Conduct capacity and airflow tests as specified in Table II. After all test apparatus and instrumentation has been connected to the heater, operate the heater for not less than 1 hour at test conditions to insure temperature stabilization and uniform operation. Following stabilization, data upon which results will be based shall be recorded every 15 minutes for a total of not less than five complete sets of data. Except for fuel consumption, results shall be based on the average of the five sets of data. Fuel consumption shall be based on the total weight of fuel consumed within the 1-hour data-recording period. Each set of data shall consist of not less than the following:

(a) Air in, °F (read at not less than four locations spaced across the return-air inlet to the heater).
(b) Air out, °F (read at not less than four locations spaced across the warm air discharge from the heater and so as not to receive radiant heat).
(c) Fuel weight, pounds.
(d) Flue-gas temperature, °F (read within 1 inch of where flue-gas analysis sample is taken).
(e) Flue-gas analysis, percent CO_2, O_2, and CO by volume (samples taken from heater exhaust stack at a point 18 inches from the heat exchanger).
(f) Static air back pressure, inches of water (read at warm air discharge opening from heater).
(g) Static and velocity air pressures, inches of water (as required to determine airflow with airflow measuring apparatus being employed).
(h) Power input characteristics, volts, watts, amps.
(i) Barometric pressure, inches of mercury.

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(j) Circulating air-blower speed, rpm (if accessible).
(k) Ambient air temperature, dry bulb and wet bulb, °F (measured at one location in test environment).

Table II. Capacity - Airflow Test Conditions

Fuel Type	External Back Pressure Inch Water	Power Input	Preproduction Heater No.				Initial Production Heater No.			
			1	2	3	4	1	2	3	4
		Type I Heater								
DF2	0.2	60 Hz	X	X	X	X	X	X	X	X
DF2	0.35	60 Hz				X				X
Gasoline	0.2	60 Hz	X	X	X	X	X	X	X	X
Gasoline	0.35	60 Hz				X				X
DF2	0.2	50 Hz				X				X
DF2	0.35	50 Hz				X				X
Gasoline	0.2	50 Hz				X				X
Gasoline	0.35	50 Hz				X				X
		Type II Heater								
DF2	0.2	400 Hz	X	X	X	X	X	X	X	X
DF2	0.35	400 Hz				X				X
Gasoline	0.2	400 Hz	X	X	X	X	X	X	X	X
Gasoline	0.35	400 Hz				X				X
		Type IV Heater								
DF2	0.2	28 VDC	X	X	X	X	X	X	X	X
DF2	0.35	28 VDC				X				X
Gasoline	0.2	28 VDC	X	X	X	X	X	X	X	X
Gasoline	0.35	28 VDC				X				X

Notes: 1. Conduct test marked "X" for each heater.
2. Conduct all tests at sea level elevation (0 - 500 feet).
3. Heater number designations (1, 2, 3, and 4) will be assigned by the Government.

4.6.2.1.1 Airflow. Airflow-measuring apparatus shall be in accordance with AMCA Bulletin No. 210. Subject to the approval of the contracting officer, a calibrated, insulated, measuring apparatus utilizing a smooth-approach nozzle may be used in lieu of the system specified in AMCA Bulletin No. 210. The special apparatus shall contain the minimum components as follows: Straightening vanes, adjustable back pressure orifice, velocity

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and static pressure tubes, thermocouple or thermometer points, airflow-measuring smooth-approach nozzle, booster fan, and plenum chamber. Readings or results obtained by means of velometers or anemometers are not acceptable.

4.6.2.1.2 Fuel. Fuel weight and consumption shall be obtained with a balance-type scale having an accuracy within 0.01 pounds.

4.6.2.1.3 Barometric pressure. Barometric pressure shall be measured with a mercury barometer and shall be corrected for temperatures of the scale and the mercury, and for the location of the barometer with regard to altitude and latitude. Readings obtained from aneroid barometers are not acceptable.

4.6.2.1.4 Temperatures. Temperatures shall be measured by appropriately located glass thermometers or thermocouples and calibrated potentiometers. Readings shall be accurate within 1/2 °F.

4.6.2.1.5 Pressures. Air pressures and pressure differentials shall be measured with liquid manometers having an accuracy of 0.01 inch of water.

4.6.2.1.6 Flue-gas analysis. Flue-gas analysis shall be made with an Orsat as prescribed in ASME Power Test Code, Supplement of Instruments and Apparatus, Part 10 - Flue Exhaust Gas Analysis or any electronic analyzer capable of producing readings of at least one hundred part per million of the various gases needed for the calculations in 4.6.2.1.7. Electronic analyzers, when used, shall be calibrated and checked with a known value of gas prior to the test. A smoke content tester as described by ASTM D 2156-65 shall be used to measure the smoke content of the flue gases.

4.6.2.1.7 Test results. The test data from the capacity test shall be used to determine the following results:

(a) Airflow in scfm calculated in accordance with AMCA Bulletin No. 210.
(b) Measured output and measured efficiency calculated as follows:

(1) $Q_1 = (\text{cfm (d)} (t_1 - t_2) (0.24) (60)$

(2) $Q_2 = W_1 \text{ X HHV}$

(3) $E_1 = \frac{Q_1}{Q_2} \text{ X } 100$

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(c) Indicated efficiency, E_2, from flue gas losses as determined for all fuels referenced in 3.14 by using flue gas temperature and flue gas analysis data in conjunction with No. 2 fuel oil curves, Figure 1, Page 14.13, ASHRAE 1977 Fundamentals Handbook, as follows:

$$E_2 = 100 \text{ minus percent flue gas losses.}$$

(d) Indicated output from measured input and indicated efficiency.

$$Q_3 = Q_2 \times \frac{E_2}{100}$$

(e) Smoke content of flue gases from flue gas analysis data.

SYMBOLS USED IN CALCULATIONS

Q_1 = measured output, Btu/hr
Q_2 = measured input to unit, Btu/hr
Q_3 = indicated output, Btu/hr
E_1 = measured efficiency, percent
E_2 = Indicated efficiency, percent
d = air density at temperature measured, lb/cu ft
W_1 = weight of fuel, lb/hr
HHV = higher heating value of fuel, Btu/lb
CO_2 = percent carbon dioxide in dry flue gases by volume
CO = percent carbon monoxide in dry flue gases by volume
O_2 = percent oxygen in dry flue gases by volume
t_1 = temperature of air leaving unit, °F.
t_2 = temperature of air entering unit, °F.
t_g = temperature of exhaust flue gases, °F.

4.6.2.1.8 Basis for failure. One or more of the following shall constitute failure of this test:

(a) Airflow. The heater delivers airflow less than specified in 3.19.
(b) Capacity. The heater has measured capacity less than specified in 3.1.
(c) Efficiency. The measured or indicated efficiency is less than specified in 3.11.
(d) Flue gas analysis. The flue gases contain amounts of smoke greater than specified in 3.17.

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4.6.2.2 Environmental.

4.6.2.2.1 Humidity. Conduct the humidity test as specified in MIL-STD-810, Method 507, Procedure I, except that the maximum temperature shall be 155° F. The heater shall not be operated during exposure. Nonconformance to 3.8.2 shall constitute failure of this test.

4.6.2.2.2 Salt-fog. Conduct the salt-fog test as specified in MIL-STD-810, Method 509, Procedure I. The heater shall not be operated during this exposure. Nonconformance to 3.8.2 shall constitute failure of this test.

4.6.2.2.3 High-temperature. Subject the heater to the following temperatures and operate as indicated:

(a) An ambient temperature of plus 155° F for 4 hours; no solar radiation.
(b) Reduce the ambient temperature to plus 80° F. Operate the heater, using gasoline, for a period of not less than 1 hour, and observe the operation.

Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.2.4 Low-temperature. Subject the heater to the following temperatures and operate as indicated:

(a) An ambient temperature of minus 65° F for 24 hours; no solar radiation. Raise the temperature to minus 50° F. After exposure and while at this temperature, start the heater using diesel fuel, Class DF-A having a cloud point not lower than minus 55° F. After heater operation has stabilized, operate the heater continuously for a period of 1 hour and record sufficient data to perform the indicated output and efficiency calculation specified in 4.6.2.1.7. Upon completion of operation on diesel fuel, change to operation on gasoline and operate for 15 minutes, Stop the heater and expose it to minus 50° F for not less than 2 hours. After this exposure, start the heater and repeat operation and recording specified above utilizing gasoline.
(b) Raise the ambient temperature to plus 80° F and examine all components for damage or deterioration. Then operate the heater, using gasoline, for a period of not less than 1 hour and observe the operation.

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One or more of the following shall constitute failure of this test:

(1) Nonconformance to 3.8.
(2) Indicated capacity and efficiency less than that obtained on the tests specified in 4.6.2.1.

Note: The storage and operational portions of the low-temperature test may be conducted separately. Perform step (b), above, following the 24 hour period at minus 65° F to complete the storage portion. Begin the operational portion by exposing the heater to minus 50° F for 4 hours prior to operation with DF-A diesel fuel.

4.6.2.2.5 Vibration. Conduct the vibration test as specified in MIL-STD-810, Method 514.2, Category F, Procedure VIII, Table 514.2-VI, Curve V. The test time shall be 60 minutes total per plane. The heater base shall remain in a horizontal plane during all vibration testing. Conduct the vibration test on a vibration machine with the heater mounted by means of four captive mounting nuts provided in the bottom. Auxiliary braces, straps, or other tiedown devices shall not be employed to secure the heater to the test apparatus. No blocking or tiedown of internal components, other than those intended to remain as part of the heater during its normal operation, shall be employed. Removable panels and access doors shall be in place during all testing. The heater shall be operated during the vibration testing. After completion of all vibration tests, operate the heater to insure that all controls and components function. Nonconformance to 3.8.1 shall constitute failure of this test.

4.6.2.2.6 Shock. Conduct the shock test with the heater mounted by means of four captive mounting nuts provided in the bottom. Auxiliary, braces, straps, or other tiedown devices shall not be employed to secure the heater to the test apparatus. No blocking or tiedown of internal components other than those intended to remain as part of the heater during its normal operation shall be employed. Subject the heater to three shocks perpendicular to each of five faces (four sides and bottom - a total of 15 shocks). Each shock shall have a peak value of 15 g's and a duration of 11-milliseconds, and shall have a wave form as required by Figure 516.2-2, MIL-STD-810. The heater shall then be started and operated for 1 hour. Nonconformance to 3.8.1 shall constitute failure of this test. No auxiliary fasteners, tiedown devices, or cushions shall be used.

4.6.2.2.7 Altitude. The heater shall be operated in a 10,000 foot altitude environment, either natural or simulated, with the ambient temperature at 70° F plus or minus 10° F. After heater operation has stabilized, operate the heater continuously for a period of 1 hour and record sufficient data to determine the indicated output, indicated efficiency, and flue gas

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smoke content as specified in 4.6.2.1.7. This test shall be conducted utilizing both DF-2 diesel fuel and gasoline, and while operating at an external back pressure of 0.24 inch water without the discharge air grille. Nonconformance to 3.1 or 3.17 shall constitute failure of this test.

4.6.2.3 Controls. Place the unit in a properly ventilated area and check visually the electrical system, fuel system, airflow system and cabinet for loose or missing parts, loose or improper connections, and electrical over-current protection and thermostat in place. Perform the pre-operational check in accordance with operating instructions. After correction of any problem area discovered above, adjust the thermostat so that the heater runs continuously. After not less than 5 minutes operation check the following:

(a) Check and adjust (as required) fuel rate for proper operation.
(b) Evidence of fuel leaks.
(c) Fuel pump and solenoid valve are operating and fuel is delivered to the burner.
(d) Ignition is on and operating properly.
(e) Combustion air blower, circulation air fan, and other components are on and operating properly.
(f) Visually observe flue-gas smoke content.

Turn switch to "off" and observe the heater for proper shutdown operation. After correction of any problem areas, turn the heater on again and check the above items again. If no additional problems are discovered, operate the unit for approximately 15 minutes before conducting the following tests.

4.6.2.3.1 Thermostat control. The thermostat shall be turned up until the heater comes on; after the heater has run for 5 minutes, the thermostat shall be turned down until the heater shuts off. The temperature differential between "on" and "off" shall be noted. Nonconformance to 3.20.1 shall constitute failure of this test.

4.6.2.3.2 High temperature safety. The heater discharge airflow shall be partially blocked during heater operation until the high temperature safety control stops the flow of fuel to the combustion chamber. After the air temperature reaches 210° F the rate of blocking shall be adjusted to limit the rate of average temperature change to 1° F per minute. The temperature of the discharge air shall be recorded when the safety control deactivates the fuel supply. Nonconformance to 3.20.3.1 shall constitute failure of this test.

4.6.2.3.3 Flame-failure safety. Disconnect the ignition wires. Start the heater with the ignition wires disconnected and record the time it takes to stop the flow of fuel to the combustion chamber. After the

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control stops the fuel flow, attempt to restart the heater by depressing the manual reset control. Nonconformance to 3.20.3.2 shall constitute failure of this test.

4.6.2.4 Durability. Operate each of three heaters for not less than 2,000 hours actual heating time under the conditions specified in 4.6.2.4.2. The heater shall be operated for not less than 250 hours at each of the power conditions specified in 3.4 for the applicable type heater. The remainder of the 2000 hour test shall be run utilizing rated power. Upon completion of the durability test, the capacity tests specified in 4.6.2.1 shall be conducted on each heater.

4.6.2.4.1 Records. The following data shall be recorded at the start of the test and every 60 hours plus or minus 10 hours thereafter:

(a) Fuel rate, pounds per hour.
(b) Flue gas temperature, °F.
(c) Flue gas analysis percent by volume of CO_2, O_2, and CO.
(d) Power input characteristics, volts, watts.

Records shall be maintained, including full explanations of all servicing and actual operating time. Maintenance actions shall not be scheduled in excess of requirements specified in 3.5 and 3.9.

4.6.2.4.2 Procedure. This test may be conducted at any ambient temperature not exceeding 80° F. The heater shall be set up on an automatic test stand designed to operate the heater at nominal fuel flow under a cyclic operation of 45 minutes on and 15 minutes off. The fuel supply shall be regulated so as to impose on the fuel pump a 7-foot suction head for 75 percent of the operational time and a 3.5-foot suction head for 25 percent of the operational time. The percentage of pump load shall be altered periodically, and no one particular suction head shall be maintained for more than 125 hours of continuous operation. Proper functioning of all controls shall be ascertained every 500 hours of total operational time. If at any time, when operating on rated electrical power input, the data specified in 4.6.2.4.1 varies by more than 10 percent from that obtained at the beginning of the test, the applicable heater components shall be given a detailed inspection to determine the cause of the variation. This test shall be conducted with the heater burning the following fuels and operating for not less than 300 hours on each fuel:

Gasoline conforming to MIL-G-3056.
Gasoline conforming to VV-G-76.
Leaded gasoline up to 100 octane grade.
Diesel fuel conforming to VV-F-800, Class DF-1.
Diesel fuel conforming to VV-P-800, Class DF-2.

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Upon completion of the 2000 hour durability test including repeat of capacity test the heater shall be completely disassembled for inspection. Parts subject to failure, wear or distortion shall be inspected and the finding recorded.

4.6.2.4.3 Bases of failure. One or more of the following shall constitute failure:

(a) Failure of a heat exchanger on any one of the heaters.
(b) More than a total of seven failures of other components which are not repetitive type failures that prevent starting or operation of the heater.
(c) Failure that is considered a safety hazard.
(d) Nonconformance to 3.9.
(e) Nonconformance to 3.4.

4.6.2.5 Sound level. The heater shall be positioned at ground level and shall be operated at rated capacity during the test. Measure the sound level at eight evenly spaced points 65 inches above ground level on a circle of 3 foot radius from the geometric center of the heater in accordance with MIL-STD-1474 for steady-state noise. Nonconformance to 3.13 shall constitute failure of this test.

4.6.2.6 Tilted-operation test. Start and operate with the bottom edge of the side of the heater raised to form a 10-degree angle from the horizontal. Repeat the procedure for each of the remaining sides. Nonconformance to 3.10 shall constitute failure of this test.

4.6.2.7 Electromagnetic interference. The heater shall be tested to determine conformance with 3.22. The contractor shall furnish the contracting officer a plan of tests and a report of tests as required by MIL-STD-461. Disapproval of the plan or the report shall constitute failure of this test.

4.6.2.8 Individual performance. Operate the heater in any ambient temperature between 50° F and 80° F for a period of not less than 30 minutes using DF-2 diesel fuel at zero static air pressure external to the heater. During this time, make adjustments to obtain proper nozzle pressure and fuel consumption rate, and perform control tests as specified in 4.6.2.3. After the above period, operate the heater under the same conditions for an additional period of not less than 15 minutes during which time record not less than the following:

(a) Fuel weight, pounds and ounces for the total 15 minute run.
(b) Flue-gas temperature, °F (read within 1 inch of where flue-gas analysis sample is taken).

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(c) Flue-gas analysis, percent CO_2, O_2, and CO by volume (samples taken from heater exhaust stack at a point 18 inches from the heat exchanger), in accordance with 4.6.2.1.6.

Immediately after completion of this test, each heater shall be completely purged of fuel.

4.6.2.8.1 Basis for failure. One or more of the following shall constitute failure of this test:

(a) Smoke in the flue gases exceeds No. 4 Bacharach.
(b) Indicated output more than 5 percent less than that obtained on the preproduction model initial capacity test.
(c) Indicated efficiency less than 70 percent.
(d) Controls failing the test specified in 4.6.2.3.

4.6.2.9 Pressure. Seal all external openings into the combustion side of the heat exchanger except one which shall be fitted with an air valve. Subject the heat exchanger to an internal air pressure of 5 pounds per square inch gage and submerge the heat exchanger in water. Any evidence of air leakage shall constitute failure of this test.

4.6.2.10 Safety evaluation. Evaluation shall be accomplished throughout testing. Nonconformance to 3.12 shall constitute failure of this test.

4.6.2.11 Human factors. The heater shall be evaluated to determine that the human factors have been met. Nonconformance to 3.29 shall constitute failure of this test.

4.6.2.12 Maintenance evaluation. Maintenance operations required during initial production tests shall be accomplished as specified in 3.5. Any assemblage or support equipment such as technical manuals, repair parts, special tools, etc., provided under the contract for operation or conducting maintenance will be evaluated. Failure of the heater to be maintained as specified in 3.5 shall constitute failure of this test.

4.6.2.13 Reliability. Using the MTBF specified in 3.30, the heater shall be tested as specified herein with accept/reject criteria in accordance with MIL-STD-781, Test Plan IIC. The heaters shall be accepted if the test time and failures fall within the "accept" region to the right of the "accept and continue test" region of Test Plan IIC, and rejected if test time and failures fall within the "reject" region. If the assessment falls within the "accept and continue test" region, additional testing as specified in 4.6.2.13.1 shall be performed until an accept/reject decision is reached.

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4.6.2.13.1 Procedure. Subject the heaters to the test specified in 4.6.2.4, using only gasoline conforming to VV-G-76, and nominal rated power.

4.6.2.13.2 Failure criteria. A failure is defined as any malfunction which the operator cannot remedy by adjustment, repair, or replacement action using the controls, parts, and tools supplied with the units within 30 minutes and which causes, or may cause:

(a) Failure to commence operation, cessation of operation, or degradation of performance below designated levels.
(b) Damage to the system or any subsystem.
(c) Personnel safety hazards.

4.7 Inspection comparison. The Government may select heaters at any time during the contract production period and subject these heaters to the examination specified in 4.6.1 and the tests marked "X" in Column 4 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production heaters. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison heater to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison heaters and shall be cause for refusal by the Government to continue acceptance of production heaters until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on heaters previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all heaters accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

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4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

120.	Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
121.	Heaters not completely drained of fuel as specified.
122.	Fuel inlet fitting not sealed as specified.
123.	Heaters not secured and cushioned within the box to prevent free movement and damage.
124.	Box closure not Method V as specified.
125.	Tools not preserved as specified.
126.	Consolidation not as specified.
127.	Gross weight exceeds 200 pounds.
128.	Strapping not as specified.
129.	Fiberboard boxes exceed the size or weight limitations.
130.	Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2), to prove prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The preproduction pack may be accomplished utilizing either the preproduction heater or a production heater. If the preproduction model heater is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the preproduction model heater and production heaters.

5.2 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Heaters. Each heater, completely drained of fuel, shall have the fuel inlet fitting sealed with a snug-fitting plastic cap. Each heater and box of consolidated components specified herein in 5.2.1.6, shall be placed in a close-fitting box conforming to PPP-B-636, W5c. The heater and box of consolidated components shall be blocked, braced, and cushioned in the box as required to prevent free movement and damage to the heater, the consolidated components, or the box. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.2.1.2 Thermostat and thermostat and power-supply receptacles. The thermostat and thermostat and power-supply receptacles for each heater shall be preserved in accordance with MIL-P-116, Method IC-2, using the contractor's carton or a close-fitting box conforming to PPP-B-566.

5.2.1.3 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.4 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A.

5.2.1.5 Technical publications. Technical publications for each heater shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3 and secured to the heater case or consolidated with the items in 5.2.1.6.

5.2.1.6 Consolidation. The thermostat, thermostat and power-supply receptacles, repair parts, and maintenance tools for each heater shall be consolidated together in a close-fitting box conforming to PPP-B-636, Domestic Class. The items shall be cushioned in the box as required to prevent free movement and damage to the items or the box. The box of consolidated items shall then be placed in the box as specified in 5.2.1.1.

5.2.2 Commercial. Each complete heater shall be preserved in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.3.1 Level A. Heaters of like description, preserved as specified in 5.2, shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, Grade B, in quantities not to exceed 200 pounds per box. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

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5.3.2 Level B. Heaters preserved as specified in 5.2 shall be packed together as specified in 5.3.1 except that boxes shall be Domestic Type and strapping shall be Finish A. As an alternate, heaters may be packed in close-fitting boxes conforming to PPP-B-640, Class 2, style optional or PPP-B-636, V3c, V11c, V13c, or V15c, style optional, in quantities not to exceed the gross weight or size limitations of the applicable box specification.

5.3.3 Commercial. Heaters preserved as specified in 5.2, shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The space heaters are intended for use in providing heated air to maintain adequate working temperatures in military vehicles or shelters. Heaters may be mounted inside a shelter in proximity to operating personnel.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of heater required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of heaters to be furnished, when applicable (see 3.3).
(f) When additional accessory panel is required (see 3.21).
(g) Color required (see 3.27).
(h) When the welding process shall not be the option of the contractor (see 3.31.6).
(i) Time frame required for submission of preproduction pack. (see 5.1).
(j) Degree of preservation and degree of packing required (see 5.2 and 5.3).
(k) When Finish A strapping is required (see 5.3.1).

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6.3 Preproduction model. Any changes or deviations of production heaters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish heaters conforming to this specification.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the heater covered by this specification as contained in the Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the heaters in accordance with this specification.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each heater.

6.7 Provisioning. The contracting officer should include provisioning requirments for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of heaters.

6.8 Supersession data. Type IV replaces Type III covered in previous editions of this specification.

6.9 Electromagnetic interference.

(a) Heaters in accordance with this specification have been previously qualified to MIL-STD-461A, Notice 4 as specified in Sections 3 and 4 of this specification. Subsequent notices and revisions should not be made applicable to a procurement contract.
(b) Bidders are cautioned against failing to properly take into consideration the radio frequency interference suppression/electromagnetic interference characteristics requirements for the end items in preparing their bids.

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.6).

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Custodians:
Army - ME
Navy - YD
Air Force - 99

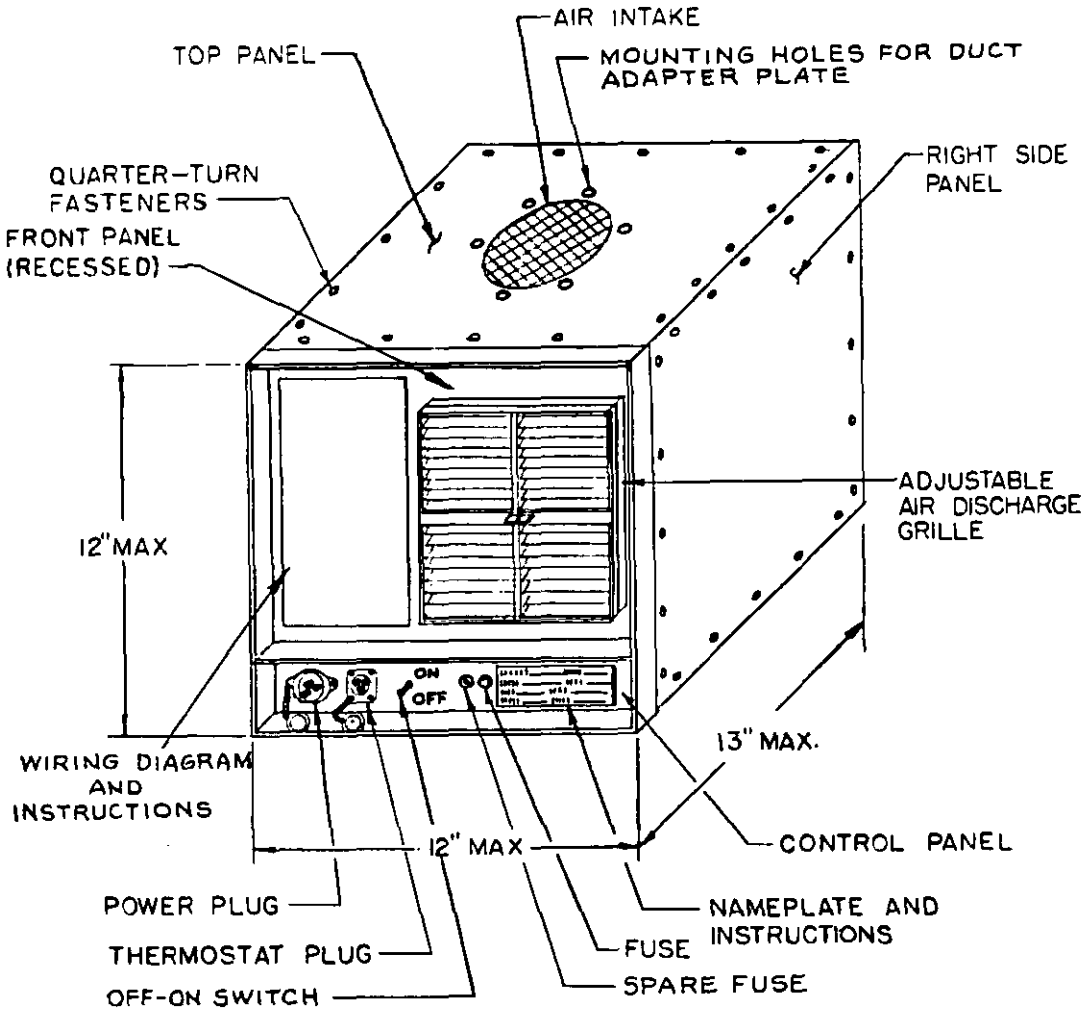
Review activities:
Army - MI
DLA - CS

User activities:
Army - EA, EL
Navy - MC

Preparing activity:
Army - ME

Project 4520-0230

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NOTE: COMPONENTS ON PANEL ARE TYPICAL ARRANGEMENT.

FIGURE 1. Heater, 15,000 Btu/hr - front view.

CX-784B

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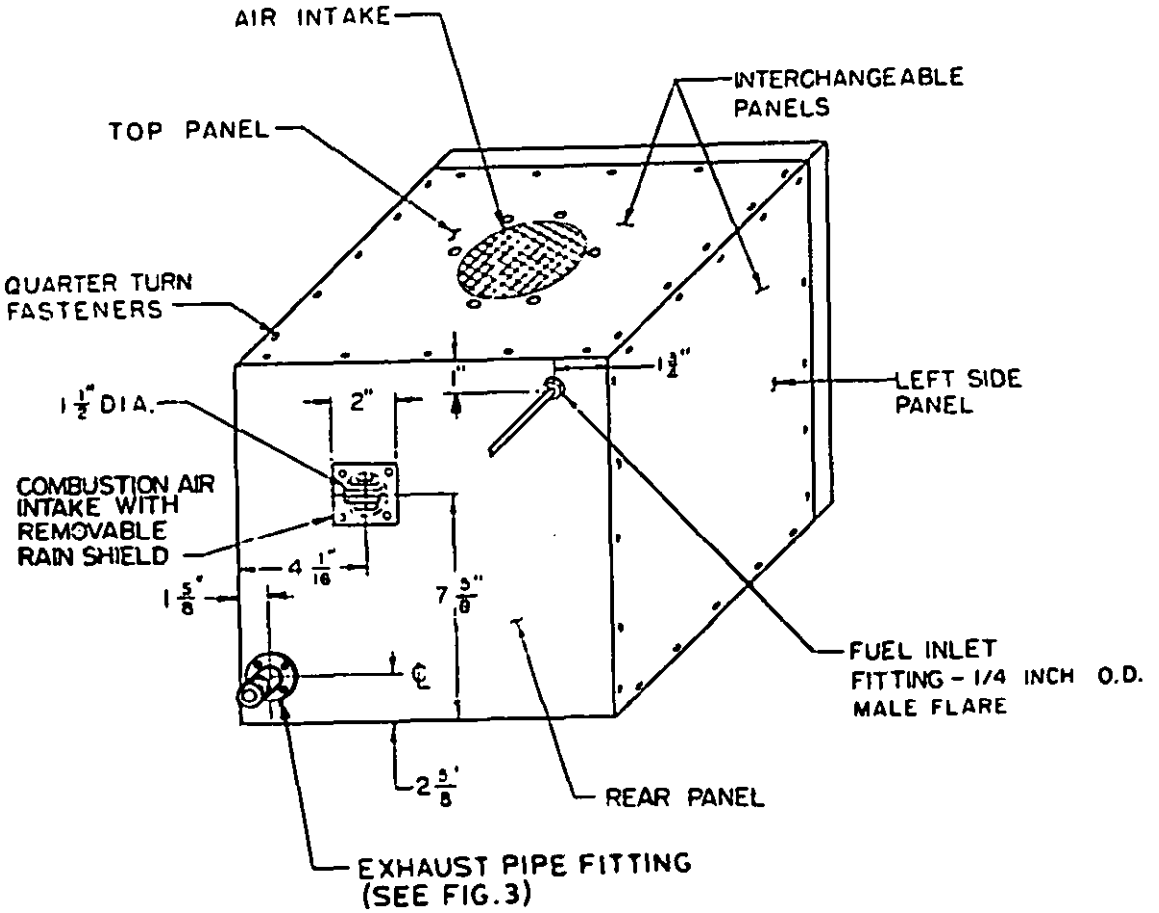
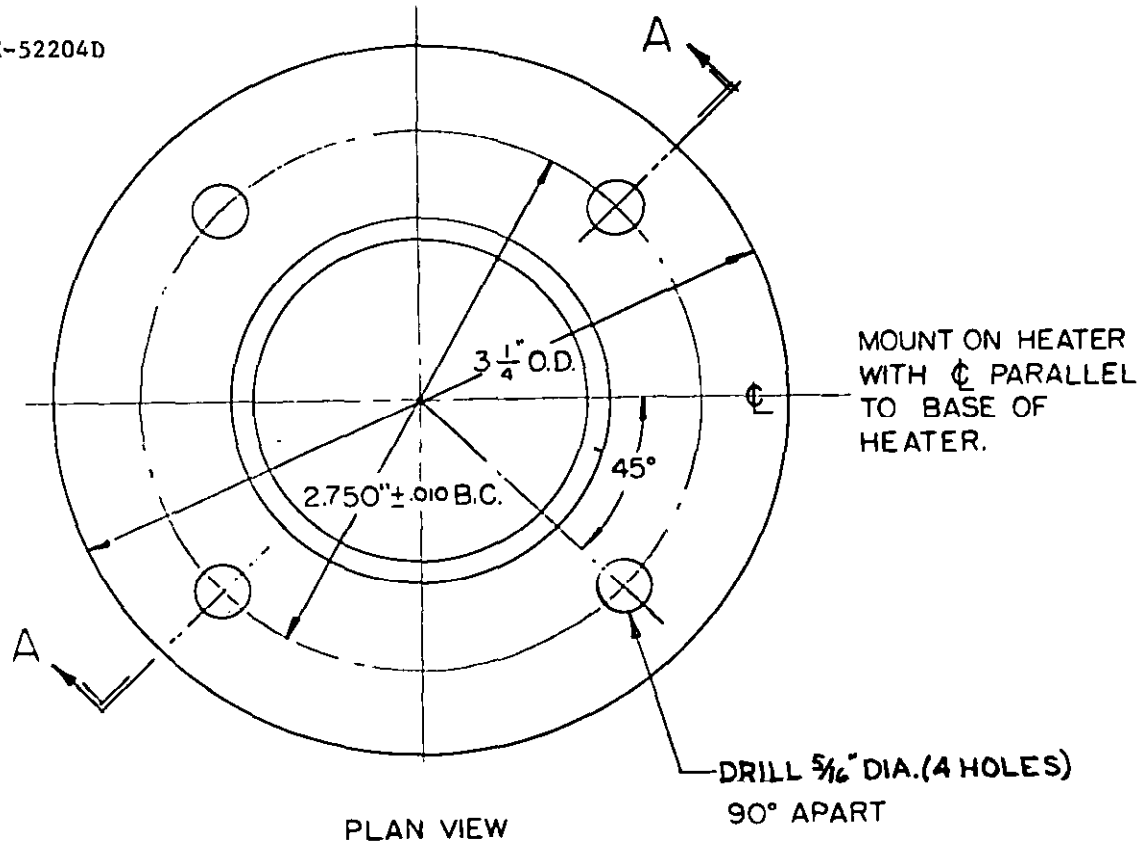


FIGURE 2. Heater 15,000 BTU/HR rear view.

X-785B

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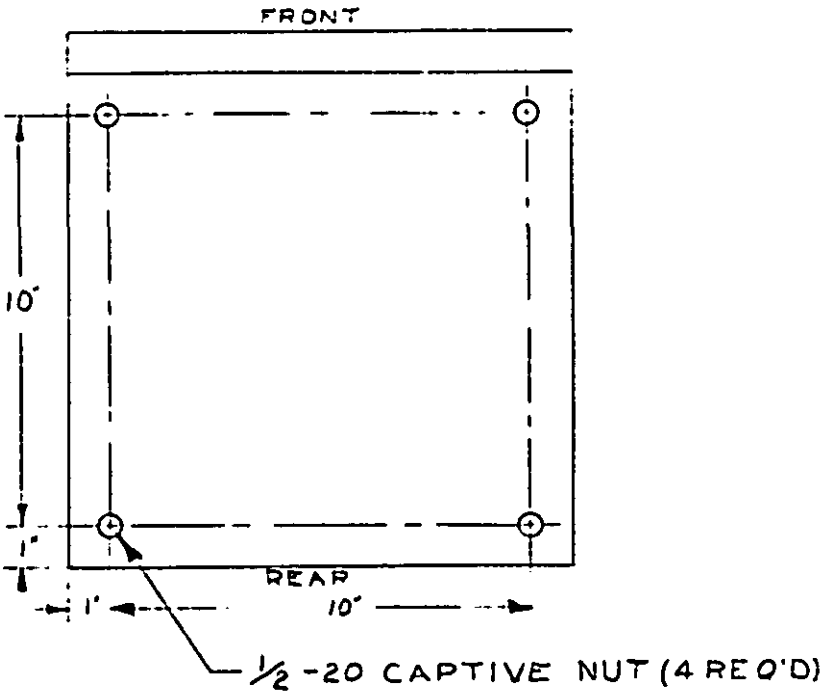


NOTE: ALL DIMENSIONS SHALL HAVE TOL OF ± 1/32 IN. EXCEPT AS OTHERWISE SPECIFIED.

FIG. 3 - HEATER, 15,000 BTU/HR - EXHAUST PIPE FITTING

X-2457

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NOTES:
1. THE CAPTIVE NUT SHALL BE REMOVABLE FROM ITS RETAINER
2. ALL DIMENSIONS SHALL HAVE TOL OF ± 1/32 INCH EXCEPT AS OTHERWISE SPECIFIED.

FIGURE 4. -HEATER, 15,000 BTU/HR - BOTTOM VIEW (MOUNTING HOLE LOCATION)

X-2460

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-H-52204D Heaters, Space: Multifuel, with Blower, 15,000 BTU/Hr

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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